

# Module

## 2

### Networking Basics

### Practical



# Training structure



EASY



SIMULATION

# Overview

- Environment: VirtualBox Lab
- VMs: Kali Linux, Windows 11, Windows Server 2019 (Domain Controller)
- Tools: Wireshark, Packet Tracer, CLI tools (`ping`, `tracert`, `ipconfig`, `nslookup`, `netstat`)

# Objectives

- Build and analyze real and simulated networks
- Observe and understand core network protocols (DHCP, ARP, DNS, HTTP/S, SMTP, SNMP, SSH)
- Perform packet analysis with Wireshark
- Design subnets and logical topologies
- Apply SOC-style traffic analysis and reporting
- Map OSI and TCP/IP models to real traffic



# Installing Tools

# Installing Network Analyzer

- Receive the network packet analyzer for your environment.
- Install as per instructions.

OR

install from <https://www.netacad.com/resources/lab-downloads?courseLang=en-US> (requires signup)

# Install wireshark

- Install wireshark on Windows 11:
  - <https://www.wireshark.org/download.html>
- Install wireshark on Kali Linux:
  - sudo apt install wireshark
  - Note: Sudo requires the user password.

# Verify installs

- Start the tools and check for error messages.

# Network Basics



# Lab 1: Physical & Virtual Networking

- Verify VirtualBox Networks:
  - `File → Preferences → Network`
  - Confirm `NAT Network (10.0.2.0/24)` and `Host-Only (192.168.56.0/24)`
- In Windows 11: `ipconfig /all`
- In Kali: `ip addr show`
- In DC01: Verify DNS and DHCP enabled

# Lab 1B: Mapping Topology

- Objective: Visualize the logical topology
- Activity:
- Create a simple diagram showing:
  - Kali → Switch → DC01 → Router → Internet
  - Label interfaces, IPs, and roles
- Tools: draw.io, PowerPoint, or Packet Tracer topology view

# Lab 2: Subnetting & Address Planning

- Objective: Calculate and assign proper subnets
- Steps:
  - Network Range: `192.168.56.0/24`
  - Create four `/26` subnets:
    - .0–.63 | .64–.127 | .128–.191 | .192–.255
  - Assign:
    - DC01 → 192.168.56.10
    - Win11 → 192.168.56.70
    - Kali → 192.168.56.130
  - Apply static IPs → verify via `ping` between all VMs

# Lab 2B: CIDR Math Practice

- Objective: Reinforce subnet and CIDR understanding

- Reference Table:

|   | A           | B                  | C                | D                   |
|---|-------------|--------------------|------------------|---------------------|
| 1 | <b>CIDR</b> | <b>Subnet Mask</b> | <b>Addresses</b> | <b>Usable Hosts</b> |
| 2 | /24         | 255.255.255.0      | 256              | 254                 |
| 3 | /25         | 255.255.255.128    | 128              | 126                 |
| 4 | /26         | 255.255.255.192    | 64               | 62                  |
| 5 | /27         | 255.255.255.224    | 32               | 30                  |
| 6 | /28         | 255.255.255.240    | 16               | 14                  |

- Assignment:

- Students design address plans for 5 departments using CIDR notation
- Must include: subnet range, broadcast, first/last usable IPs

# Lab 3: DHCP & Static IP

## Configuration 1

- Objective: Configure DHCP and verify leases
- Steps:
  - On DC01 → DHCP Manager → Create new scope:
    - Range: `192.168.56.100–192.168.56.200`
    - Subnet mask: 255.255.255.0
    - Default gateway: 192.168.56.1
    - DNS: 192.168.56.10
  - On Win11 → set to automatic IP assignment
  - Run `ipconfig /release` and `ipconfig /renew`
  - Capture DORA in Wireshark using filter: `bootp or dhcp`

# Lab 3: DHCP & Static IP

## Configuration 2

- DORA Process:
  - Discover: Client broadcasts request
  - Offer: Server offers IP address
  - Request: Client accepts offer
  - Acknowledge: Server confirms lease

# Lab 4: ARP & ICMP Analysis

- Objective: Observe local network communication
- Steps:
  - In Kali → `ping dc01.lab.local`
  - Capture in Wireshark → filter: `arp or icmp`
  - Identify ARP "Who has..." and ICMP echo requests
  - Observe MAC-to-IP mapping changes
  - Check ARP cache: `arp -a` (Windows) or `ip neigh` (Linux)

# Lab 4B: TCP/IP vs OSI Model Mapping

- Objective: Compare both models practically

- Model

| OSI Layer        | TCP/IP Layer   | Protocol Examples    | Where to Observe                                                       |
|------------------|----------------|----------------------|------------------------------------------------------------------------|
| 7 (Application)  | Application    | HTTP, DNS, SMTP, SSH | Wireshark: Follow TCP Stream, decoded protocol data                    |
| 6 (Presentation) | Application    | TLS, encryption      | Wireshark: TLS handshake, certificate exchange                         |
| 5 (Session)      | Application    | Session tokens       | Wireshark: HTTP headers (Set-Cookie, Session-ID)                       |
| 4 (Transport)    | Transport      | TCP, UDP             | Wireshark: TCP 3-way handshake (SYN/ACK), port numbers, netstat output |
| 3 (Network)      | Internet       | IP, ICMP, ARP        | Wireshark: IP header, ping replies, tracer hops, routing tables        |
| 2 (Data Link)    | Network Access | Ethernet             | Wireshark: Frame details, source/destination MAC, arp -a output        |
| 1 (Physical)     | Network Access | Cables, signals      | Packet Tracer: cable selection, link lights, physical topology view    |

- Activity:
  - Open previous Wireshark captures
  - Annotate packets with both OSI and TCP/IP layer labels
  - Identify where layers collapse in TCP/IP model

# DNS & HTTP/HTTPS Traffic

- Analyze application-layer communication
- Part A – DNS
  - In Kali: ``nslookup google.com``
  - Filter in Wireshark: ``dns``
  - Identify query and response packets
  - Note DNS uses UDP port 53 (usually)

# DNS & HTTP/HTTPS Traffic

- Analyze application-layer communication
- Part B – HTTP/HTTPS
  - In Kali: ``curl -v http://example.com`` and ``curl -v https://example.com``
  - Compare captured traffic (``http`` or ``tls``)
  - Discuss visible vs encrypted content
  - Identify TLS handshake process

# Lab 5B: SMTP Traffic Analysis 1

- Objective: Analyze email protocol security (read both slides first!)
- Steps:
  - Configure simple SMTP test:
    - Use public SMTP test server or configure on DC01
  - From Win11 PowerShell:
    - `Send-MailMessage -To "test@example.com" -From "student@lab.local" -Subject "Test" -Body "Testing SMTP" -SmtpServer 192.168.56.10`

# Lab 5B: SMTP Traffic Analysis 2

- Objective: Analyze email protocol security
- Steps:
  - Capture in Wireshark: filter `smtp or tcp.port == 25`
  - Analyze conversation:
    - HELO/EHLO greeting
    - MAIL FROM / RCPT TO commands
    - DATA section (message body)
  - Identify plaintext credentials if present

# Lab 5B: SMTP Traffic Analysis 3

- Compare:
  - SMTP (port 25): Unencrypted
  - SMTPS (port 465): SSL/TLS encrypted
  - Submission (port 587): STARTTLS encrypted
- Security Discussion:
  - Why email is easily intercepted
  - Importance of encryption for sensitive data

# Lab 6: SSH and Secure Remote Access 1

- Objective: Use secure shell for remote management
- Steps:
  - On DC01 → Enable OpenSSH Server feature:
    - Settings → Apps → Optional Features → Add OpenSSH Server
  - On Kali → ``ssh administrator@192.168.56.10``
  - Observe key exchange and authentication process
  - Capture traffic in Wireshark with filter: ``tcp.port == 22``

# Lab 6: SSH and Secure Remote Access 2

- Key Observations:
  - Encrypted payload (no readable data)
  - Key exchange protocol
  - Authentication methods
- Compare to Telnet:
  - SSH: Encrypted (secure)
  - Telnet: Plaintext (insecure, deprecated)

# Lab 6B: SNMP Monitoring 1

- Objective: Query network device statistics
- Part A – Packet Tracer SNMP:
  - In Packet Tracer: Add router and enable SNMP
    - Router(config)# snmp-server community public RO
    - Router(config)# snmp-server location Lab
    - Router(config)# snmp-server contact admin@lab.local
  - Note router IP address

# Lab 6B: SNMP Monitoring 2

- Part B – SNMP Query from Kali:
  - Install SNMP tools: ``sudo apt install snmp snmp-mibs-downloader``
  - Query device: ``snmpwalk -v2c -c public [ROUTER_IP]``
  - Specific queries:
    - System info: ``snmpwalk -v2c -c public [IP] system``
    - Interfaces: ``snmpwalk -v2c -c public [IP] interfaces``

# Lab 6B: SNMP Monitoring 3

- Part C – Capture Analysis:
  - Capture SNMP traffic in Wireshark (filter: `snmp`)
  - Identify:
    - Community string (security risk if `"public"`)
    - MIB objects queried
    - Response data
- Security Discussion:
  - SNMPv1/v2c: Plaintext community strings
  - SNMPv3: Encryption and authentication
  - Community strings = passwords (should be complex)

# Lab 7: Diagnostic Tools (Windows)

- Objective: Practice network troubleshooting commands

- Lab Activity:
  - Run each command
  - Document outputs
  - Identify local gateway and DNS server
  - Check active connections

| Command           | Function            | Example                | Output                        |
|-------------------|---------------------|------------------------|-------------------------------|
| ping              | Test connectivity   | ping dc01.lab.local    | Reply/timeout, latency        |
| tracert           | Show hop path       | tracert 8.8.8.8        | Each router hop               |
| ipconfig/all      | Show network config | Full adapter details   | IP, MAC, DNS, DHCP            |
| ipconfig/release  | Release DHCP lease  | Forces IP release      | Confirmation                  |
| ipconfig/renew    | Renew DHCP lease    | Gets new IP            | New IP assigned               |
| ipconfig/flushdns | Clear DNS cache     | Removes cached entries | Confirmation                  |
| nslookup          | Resolve domains     | nslookup lab.local     | IP address                    |
| netstat -ano      | Show connections    | Lists all connections  | Local/remote IPs, ports, PIDs |
| route print       | Show routing table  | Display routes         | Network destinations          |
| arp -a            | Show ARP cache      | MAC-IP mappings        | Neighbor devices              |

# Lab 7B: Diagnostic Tools (Linux)

- Objective: Run equivalent tools in Kali

- Lab Activity:

- Run each command
- Document outputs
- Identify local gateway and DNS server
- Check active connections

| Command        | Function           | Example                     |
|----------------|--------------------|-----------------------------|
| ping -c 4      | Check latency      | Limited to 4 packets        |
| tracert        | Show route         | Similar to tracert          |
| ip addr show   | Interface config   | Replaces ifconfig           |
| ip route show  | Show routing table | Display routes              |
| ip neigh       | Show ARP cache     | MAC-IP mappings             |
| netstat -tulpn | List services      | Shows listening ports       |
| ss -tulpn      | Modern netstat     | Faster alternative          |
| dig            | DNS query          | More detailed than nslookup |
| nslookup       | DNS lookup         | Cross-platform              |
| host           | DNS lookup         | Simple queries              |

# Wire Shark



# Lab 8-9: Wireshark Deep Dive

- Follow handbook for detailed instructions
- Goals:
  - Layered filtering and pattern recognition
  - Understanding reconstruction and forensics
  - Understand transport layer protocol selection
  - Identify unusual or dominant traffic types
- Discussion points:
  - Forensic applications (incident response)
  - Privacy implications (unencrypted traffic)
  - Why HTTPS prevents this (encryption)
  - Ethical and legal considerations

# Packet Tracer



# Lab 10-11: Packet Tracer basics

- Follow handbook for detailed instructions
- Goals:
  - Create tangible topology using OSI layers
  - Visualize routing at OSI Layer 3
  - CLI familiarity with switching
  - Reinforce encapsulation process
  - Link networking to basic security policy
- Discussion points:
  - - ACL processing: top-down, first match wins
  - - Implicit deny all at end
  - - Importance of order
  - - Inbound vs outbound application

# Incident Investigations



# Lab 12-13: Incident analysis

- Follow handbook for detailed instructions
- Goals:
  - Develop SOC-style analytical workflow
  - Documentation & communication of technical results
- Deliverables:
  - Incident investigation report
  - Baseline report

# Summary of Key Skills



# Key skills Acquired 1

- Technical Skills Acquired:
  - Network mapping and subnetting (CIDR notation)
  - Protocol analysis: DHCP, ARP, DNS, HTTP/HTTPS, SMTP, SNMP, SSH, ICMP
  - CLI diagnostics (Windows/Linux)
  - Wireshark filtering and forensics
  - Packet Tracer simulation (routing, switching, ACLs)
  - Incident investigation and reporting
  - Traffic baselining

# Key skills Acquired 2

- OSI Model Mastery:
  - Layer 1 (Physical): Cable types, PoE
  - Layer 2 (Data Link): Ethernet, MAC addresses, ARP, switches
  - Layer 3 (Network): IP addressing, routing, ICMP
  - Layer 4 (Transport): TCP vs UDP, ports, connections
  - Layer 5-7 (Session/Presentation/Application): HTTP, DNS, SSH, SMTP, SNMP

# Key skills Acquired 3

- SOC Analyst Skills:
  - Packet capture and analysis
  - Protocol identification
  - Anomaly detection
  - Incident documentation
  - Baseline creation

# Self-Study Recommendations

- Practical
  - - Practice Wireshark on home network (your own traffic only!)
  - - Build more complex Packet Tracer topologies
  - - Research CVE database for network vulnerabilities
  - - Follow security blogs (e.g., Krebs on Security, Threat Post)
  - - Malware Traffic Analysis: <https://malware-traffic-analysis.net/>
  - - Wireshark Sample Captures: <https://wiki.wireshark.org/SampleCaptures>
  - - Packet Tracer Labs: <https://www.packettracernetwork.com/>
- Reading
  - Wireshark User Guide: <https://www.wireshark.org/docs/>
  - Packet Tracer Tutorials: Cisco NetAcad
  - IANA Port Numbers: <https://www.iana.org/assignments/service-names-port-numbers/>

# Thank you